

# Cutting Edge Solutions: Development of a Plastic Shredder

Steven Baugh, Robert Hestand, Jasmine Koski, Bridget Lannigan, Victor Levey



## Project Statement

- We developed and documented a plastic shredder for use in the PSU Electronics Prototyping Lab. The project serves as the first step in a budget pelletizer recycling process.

## Motivation

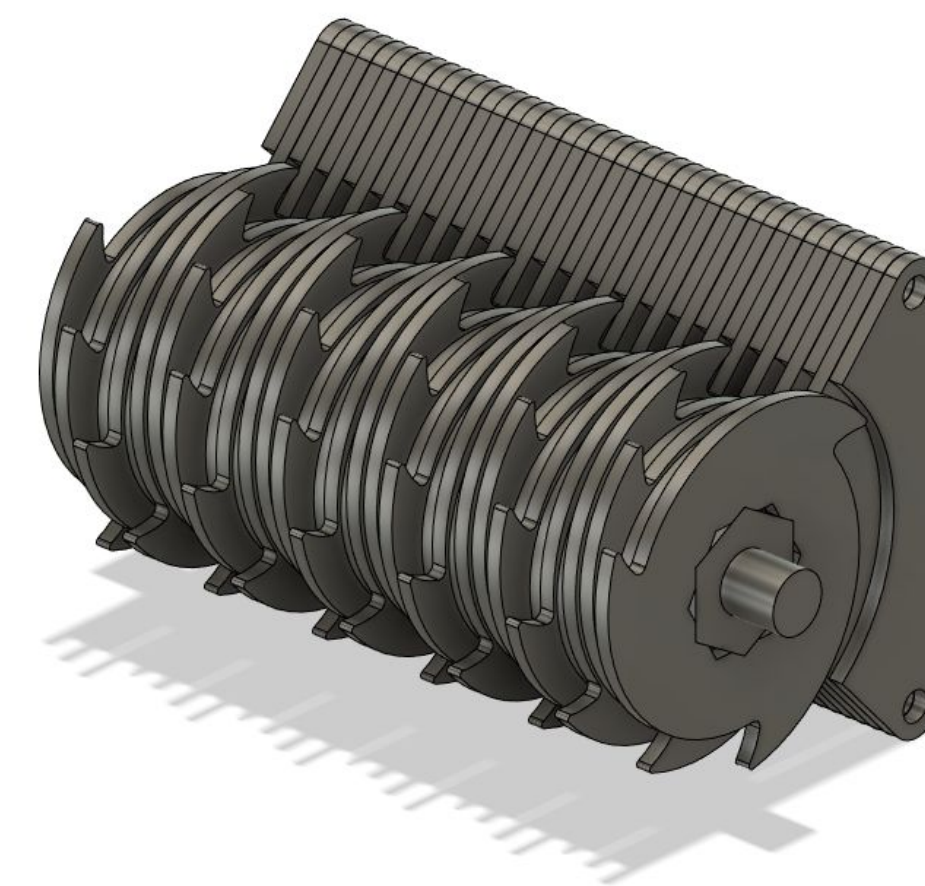
- Further Existing Research
- Improved Tooth Design
- Lower Cost
- Improved Accessibility

## Key Requirements

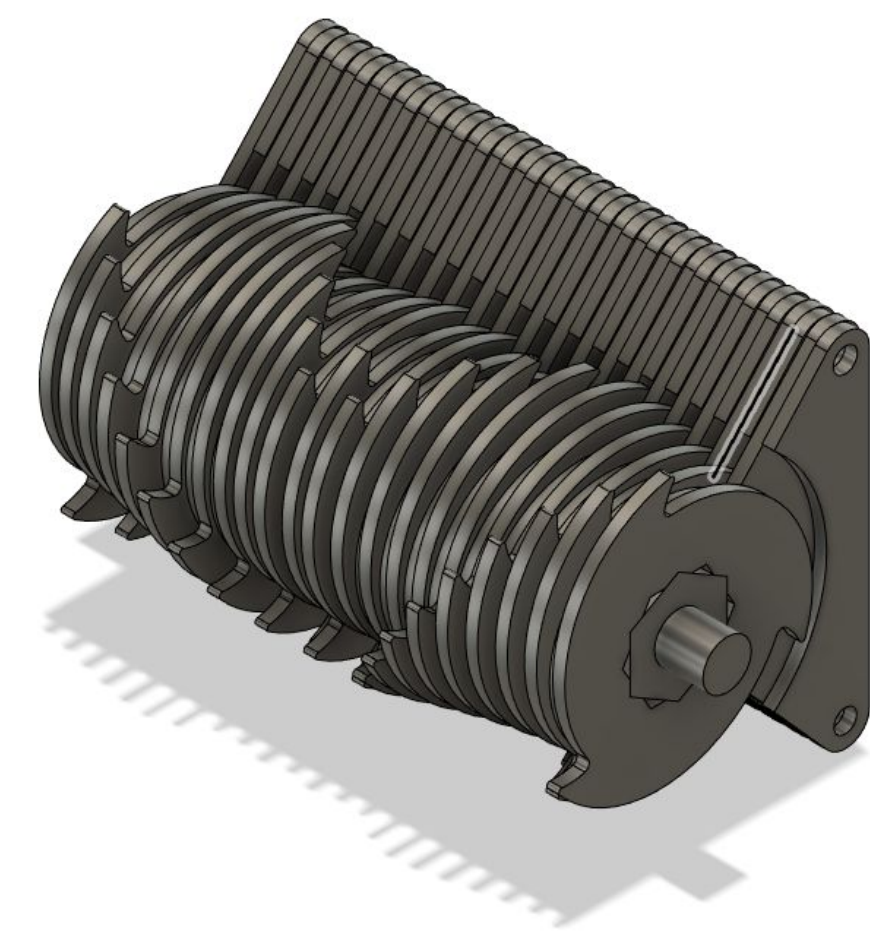
- Shred Plastic Consistently
- Laundry Container Input Size
- Maintainability
- Lower Cost than Competitors
- VFD Utilization
- Run in Forward and Reverse
- Have an Emergency Stop
- Two-hand Operation for Safety
- One Step Lock Out Procedure

## Shredder Design Concepts

The final version of our shredder utilizes a 4" normal diameter on the shredding teeth and feature 3 teeth per cutter.

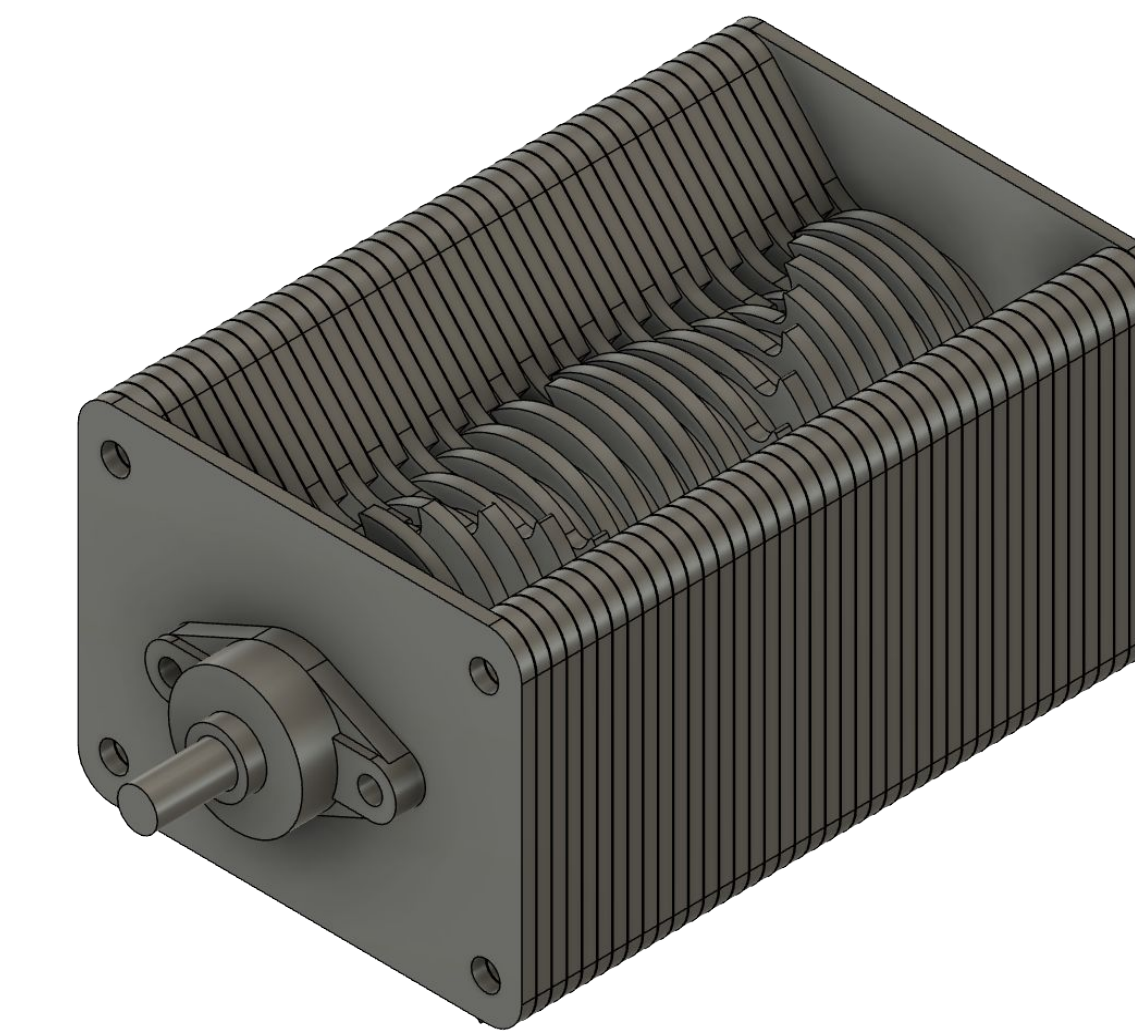


A hexagonal shaft was chosen both for its accessibility and for the ability to easily index the shredder teeth.



## Final Design

The stationary cutters are incorporated to provide a method of shearing the plastic.



The shredding teeth are able to be oriented a variety of ways for versatility and experimentation.



## Electrical Components

### What is a VFD?

A Variable Frequency Drive is a motor controller that adjusts speed by varying the frequency and voltage supplied.



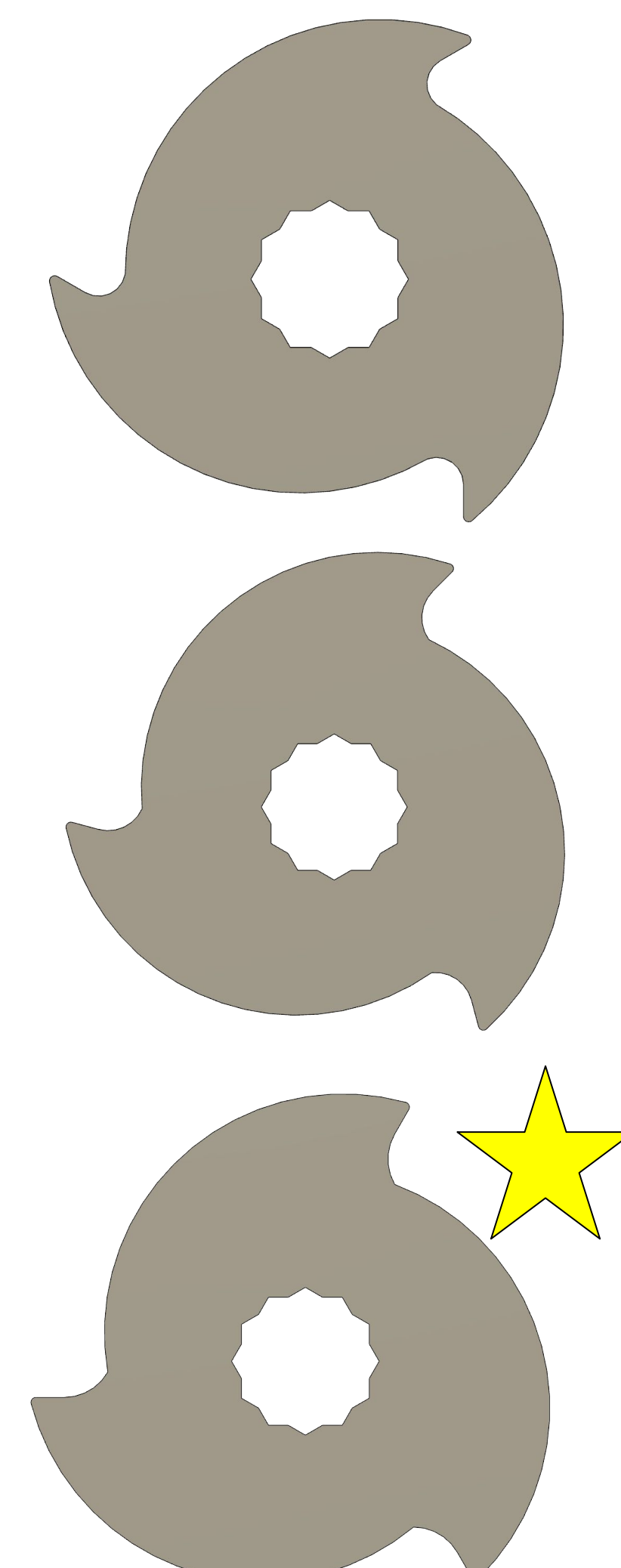
### Requirements Met:

- VFD works
- Two run buttons for operation. Both buttons must be held down for motor to energize. Motor will stop if a run button is released, or the stop or E-Stop is pressed.
- Forward and Reverse work.
- One Step Lockout at Electrical Plug



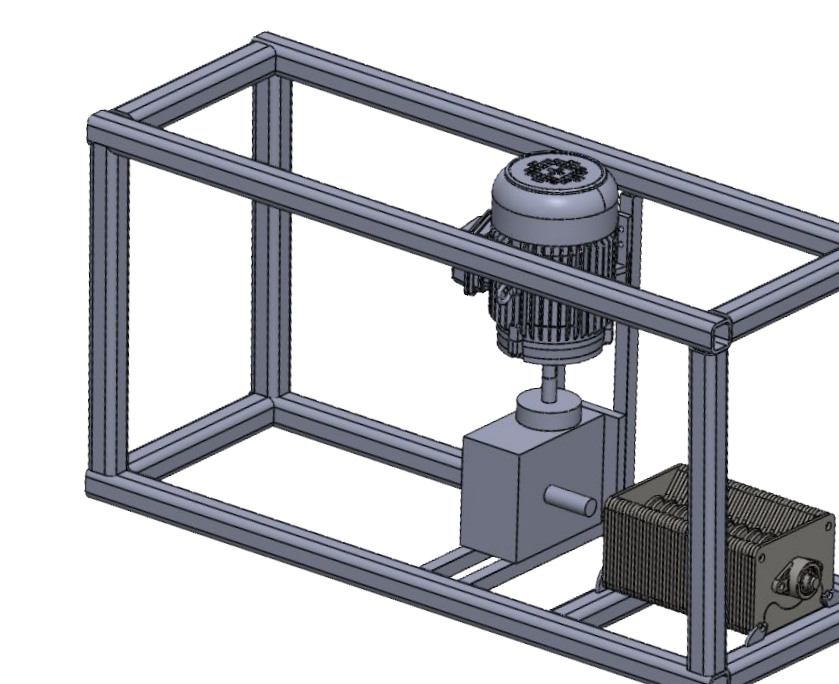
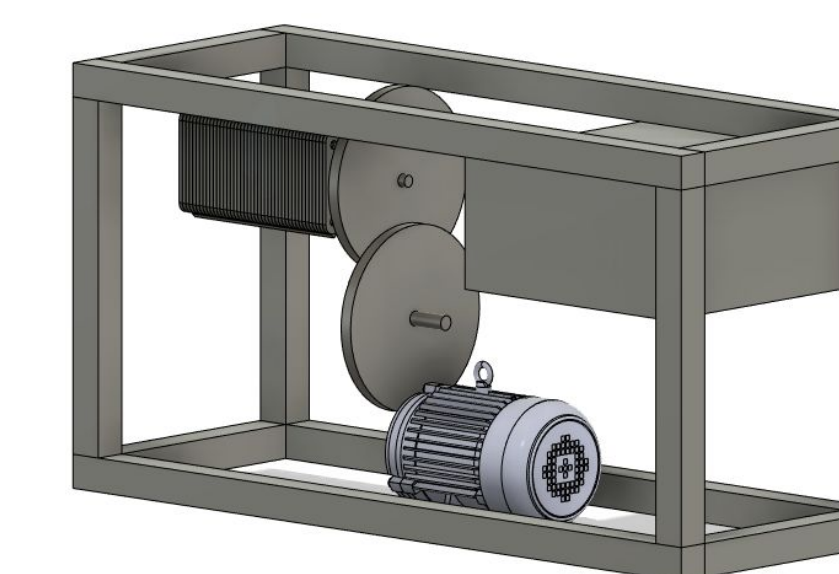
## And the Winner is...

We examined shredder teeth designs with 30°, 45°, and 60° teeth. Through experimental design, the 60° cut tooth required the least amount of shredding force. This resulted in the 60° cut tooth being picked for the final assembly.



## Finding the Right Speed

We examined both the use of pulleys and the use of a gearbox to utilize in conjunction with the VFD. Ultimately, we decided on the use of a gearbox for simplicity, cost, and ease of use for the end user.



## Special Thanks

Our team would like to extend thank yous to:

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- Ed Ivory, our client and EPL program manager, for the consistent feedback and aid. Ed was an excellent capstone client.
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